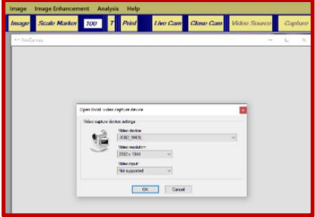
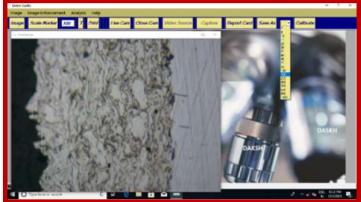


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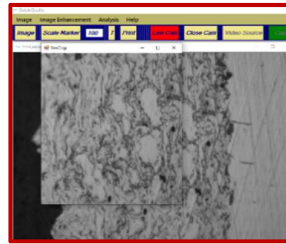
Operations/Work/Job Activity covered by this assessment: Optical Micro Scope	
Required Personal Protective Equipment (PPE) :	
 Safety glasses must be worn at times.	 Long and loose hair must be contained.
 Appropriate footwear with substantial uppers must be worn.	 Close fitting/protective clothing to cover arms and legs must be worn.
 Respiratory protection devices required.	 Oil free leather gloves and spats must be worn.
	 Hearing protection must be used when using this machine.
1. Open Image analysis software from the Desktop [Fig1] 2. Switch on the POWER to optical Microscope[Fig 2] 3. Place the mounted sample on the stage with sample facing downwards[Fig 2] 4. Click on `Video Source` in the Image analysis software displayed on the screen [Fig 3] 5. Select USB2_5M(1) in the video capture device window and set resolution at 2592X1544 [Fig 3] 6. Adjust the rough and fine focusing Knob in the optical microscope to get a well-focused image on the screen 7. Select the magnification lens of 20X in the objective and calibrate for 200 X mag.[Fig 4] 8. Select Capture and save the image with the micron marker embedded in the image as shown in Fig 5 9. Select ROI from Image enhancement drop down menu [Fig 6]	
	 [Fig1] 
	 Fig.3
 Fig.6	 Fig.5
	 Fig.6

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10. Select area of interest for porosity Analysis [Fig 7a and 7b]



[Fig 7a]



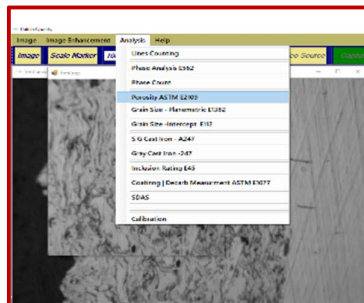
[Fig 7b]

11. Select **Porosity ASTM 2109** in Analysis Module

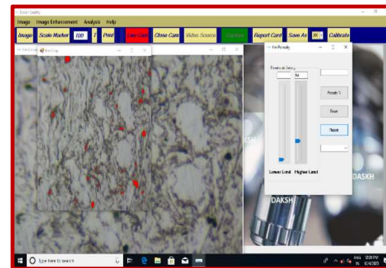
[Fig 8a] and a window appears as shown in [Fig 8b]

12. For porosity measurement, in threshold setting, keep the lower limit at minimum and vary the higher limit such that all pitch dark locations in the microstructure corresponding to porosity are selected.

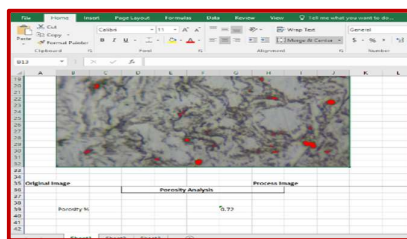
13. Click 'Porosity' button to determine the porosity percentage in the area selected and then click Report 'Report' [Fig 7b] to generate report along with microstructure of selected area as shown in **Fig 8c**. The entire process is repeated for 10 or more fields as required on the sample.



[Fig 8a]



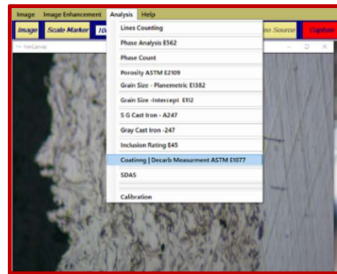
[Fig 8b]



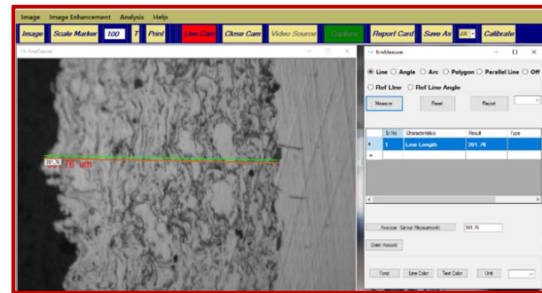
[Fig 8c]

14. For thickness measurement of coatings, select 'Coating ASTM E1077' in Analysis module [Fig 9a] and a window appears as shown in **Fig 9b**. Move Cursor to the starting point of the coating and drag it to the end of the coating at interface, then select Measure in the window, thickness measured will appear in the Table underneath. Repeat the exercise for five different locations for thickness measurement. Then select "Report" and save the measured thickness data and the average obtained in an excel file. The entire process is repeated for 10 or more fields as required on the sample.

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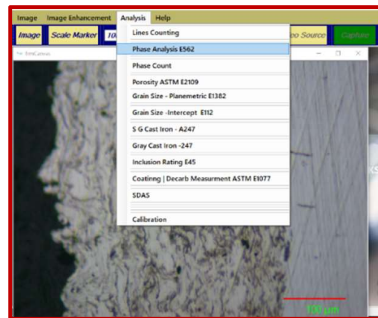


[Fig 9a]

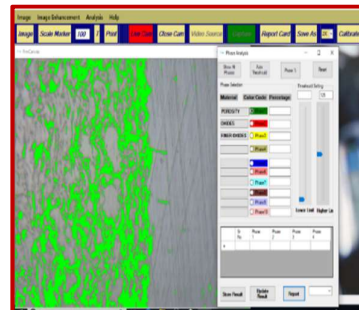


[Fig 9b]

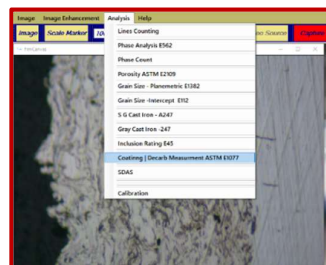
15. For determining the percentage of oxides in the coating, select “Phase analysis E562” in the Analysis Module [Fig 10a] and a window as shown in the right [Fig 10b]. Select Phase 1 and measure porosity as per the procedure shown above in S.No12. Then select Phase 2 [Red] and adjust the both lower and upper limits of threshold setting, so that the identified oxides are selected by red colour in the microstructure. Repeat the process for Phase 2 so that finer oxides if any are identified and selected. Press the phase % button followed by Report for generating detailed report with different phases in the microstructure identified and the Percentage of each phases. The entire process is repeated for 10 or more fields as required on the sample.



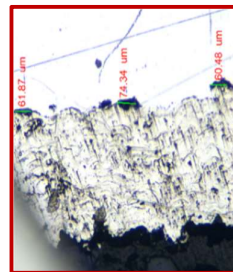
[Fig 10a]



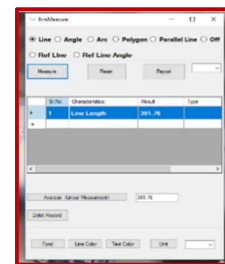
[Fig 10b]



[Fig 10a]



[Fig 10b]



[Fig 10c]

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16 For determining interfacial contamination (IC) of the coatings, select 'Coating ASTM E1077' in Analysis module **[Fig 10a]** and a window appears as shown in Fig 10b. Move Cursor to the point of interest i.e., grid contamination at interface appearing as black in color and measure its length which then appears in the Table. Repeat the exercise for other interface contaminants for its length measurement. Then select "Report" and save the measured thickness data and the average obtained in an excel file. The entire process is repeated for 10 or more fields as required on the sample.

Measure the length of the field **L** selected for measuring interfacial contamination

If the number of fields measured is 20, then total length scanned is **20 x L** microns.

Let average of Interfacial contamination in various fields are recorded as IC1,IC2.....IC20

Total length of interfacial contamination = IC1+IC2+IC3+.....+IC20 = **IC**

Interfacial contamination % = Total interfacial contamination/Total coating length.

Total Interfacial contamination % = **IC%*100**

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